Corian

Corian is a brand of solid surface material created by DuPont. Its primary use is as a countertop, benchtop surface, wash basin, and wall panel. Though it has many other applications. It is composed of acrylic polymer and alumina trihydrate (ATH), a material derived from bauxite ore. Corian is the original material of this type, created by DuPont scientists in 1967.[1] Since DuPont's patent on solid surfaces ran out, a number of direct competitors to Corian have come out.

Corian brand logo

Corian is manufactured in three thicknesses: 6 millimetres (0.24 in), 12 millimetres (0.47 in), and 19 millimetres (0.75 in). Most Corian is manufactured at a DuPont facility near Buffalo, New York.[2] Cross-section cuts show consistent color and particulate patterning evenly distributed throughout the material, giving rise to the category name "solid surface".

The former Corian logo

Corian must be sold and installed by a DuPont certified fabricator; such installations come with a 10-year warranty covering both the product and installation, for interior residential applications.[3]

History

Donald Slocum, a DuPont chemist, is credited with inventing Corian solid surface in 1967.[4] His name appears on the patent issued in October 1968.[5] The product was first introduced for sale in 1971, at the National Association of Home Builders meeting in Houston, Texas.[1]


An integrated Corian sink

Originally conceived as a kitchen and bath material available in a single color, Corian is manufactured and delivered in more than 100 colors.[6]

In 2013, the company announced its Endless Evolution initiative in an effort to improve the material and find additional applications for its use.[7] As part of this initiative, in 2014 DuPont introduced its "Deep Color" technology.[8] The enhancement allows for the material to be created in deeper, darker colors that are more resistant to scratches and cuts than earlier generation Corian material.[9]

In 2017, Corian marked its fiftieth year with a new look and marketing campaign.[10] Its renewed visual identity and logo was designed and developed by Italian branding agency GBR Design.[11][12]

Product lines

DuPont has issued various sub-branded releases of the material which contain unique design elements or methods of manufacture. Notably these have included:

- Corian Private Collection: First colors introduced in 2002; this product line is inspired by the randomness of patterns found in nature. Some colors and patterns in this product line resemble stone and other natural materials and all colors have complex non-repeating patterns not found in other Corian products.[13]
- Corian Terra Collection: First colors introduced in the 2000s; this product line contains between six and twenty percent recycled content.[14]
- Corian Illumination Series: First colors introduced in 2007; this product line is semi-translucent allowing for new designs calling for backlit applications.[15]
- Introduced in 2010, the Corian Metallic Series features gold and silver flecks for a dynamic, metallic look. This unique composition offers a sense of depth, creating a visual effect of changing colors when viewed from different angles.
- Martha Stewart Living Collection: First colors introduced in the United States in 2010; this product line is a collection of colors created by home improvement celebrity Martha Stewart.[16]
- Corian Deep Color Technology: First colors introduced in 2013; this product line uses new "Deep Color" technology to produce darker, more scratch-resistant colors.
- Corian Quartz: Rebranded in 2018, DuPont's Zodiaq quartz product was re-introduced as Corian Quartz. This product is an engineered stone and is not a solid surface material in a technical sense.

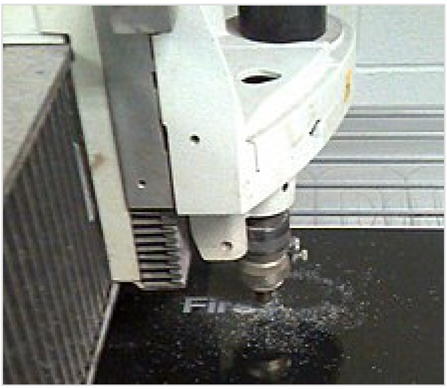

Corian samples


Martha Stewart Living Collection samples

Material characteristics

Corian consists mainly of aluminum trihydroxide (55-60%) and polymethyl methacrylate (34-45%) with trace elements of iron oxide black, carbon black, titanium dioxide, colorants and methyl methacrylate.[17] Its characteristics includes:

- Non-porous
- Stain resistant
- Seamless appearance: In the fabrication process, joints can be made nearly invisible by joining adjacent pieces with Corian's own color-matched two-part acrylic adhesive. The joined edge can be sanded and polished to create a visually imperceptible joint. This seamless appearance is a signature characteristic of the material.[18]
- Repairable and renewable: Cuts and scratches can be abraided then further buffed to restore original surface finish.
- Thermoformable: Flexible when heated, Corian can be shaped and molded into generally limitless forms which can be used in commercial and artistic projects through a process called thermoforming.


Corian being engraved for signage

Heat resistance: the material is heat resistant up to 100 °C (212 °F), but can be damaged by excess heat. DuPont recommends the use of trivets when the material is installed in kitchens.[19]

Scratches: The material can be scratched, with scratches particularly noticeable on darker colors.

Corian does not lose its visual appearance or fade for many years, sometime decades.

"Walking" table made of thermoformed Corian

Safety

Safety of installed material

Corian meets or exceeds current emissions guidelines for volatile organic compounds (VOCs), hazardous air pollutants (HAPs) and is "Greenguard Indoor Air Quality Certified". Corian is nontoxic and nonallergenic to humans. It is free of heavy metals and complies with the EU Directive 2002/95EC on the Restriction of Hazardous Substances (RoHS). Its hygienic properties make it popular in installations where maintaining sanitary conditions is important (e.g. hospitals and restaurants).[20]

Fabricator safety

In 2014, the *New England Journal of Medicine* reported a case of a 64-year-old exercise physiologist who died from lung disease consistent with idiopathic pulmonary fibrosis after 16 years of exposure to Corian dust. Dust from Corian was found in the patient's shop of Corian fabrication and lung upon autopsy. The authors said that the case was consistent with Corian dust causing idiopathic pulmonary fibrosis, but did not prove causality.[21] DuPont scientists responded that exposure to other materials could not be ruled out, nor did they rule out it was not caused from the dust which consists of aluminum trihydrate (ATH) derived from bauxite. Fabricators must properly protect themselves from fine Corian particulates generated during milling, and sanding. Fabricators should always wear a proper certified respirator and keep shop and environment clean of Corian dust at all times.

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External links

- Official website (https://www.corian.com/)